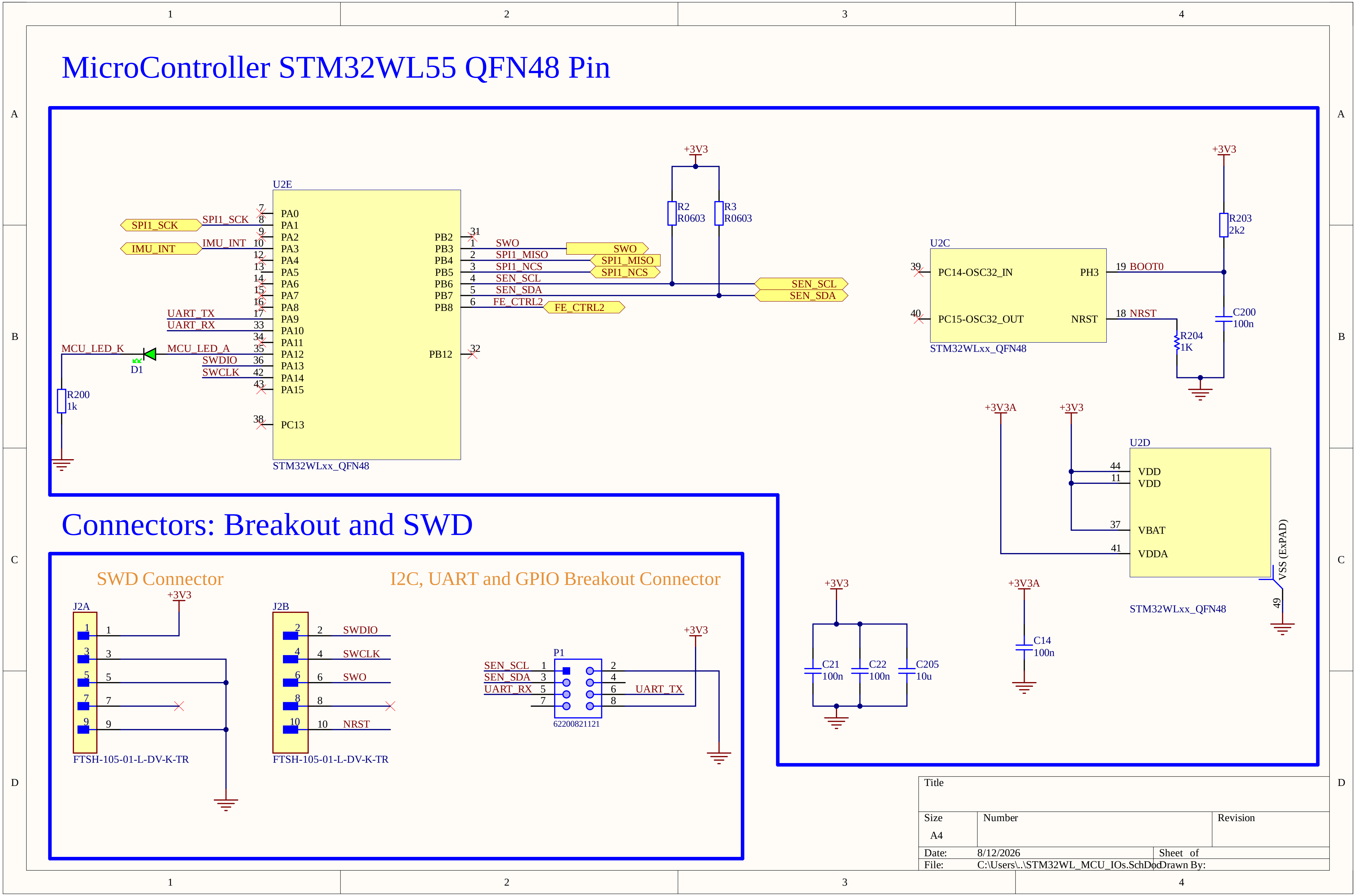

MicroController STM32WL55 QFN48 Pin
U2E
STM32WLxx_QFN48
PA0 PA1 PA2 PA3 PA4 PA5 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15 PC13
PB2 PB3 PB4 PB5 PB6 PB7 PB8 PB12
SPI1_SCK
IMU_INT
UART_TX
UART_RX
MCU_LED_K
MCU_LED_A
SWDIO
SWCLK
D1
R200
1k
SWO
SPI1_MISO
SPI1_NCS
SEN_SCL
SEN_SDA
FE_CTRL2
+3V3
R2
R0603
R3
R0603
U2C
PC14-OSC32_IN
PC15-OSC32_OUT
PH3
NRST
BOOT0
R203
2k2
R204
1K
C200
100n
+3V3A
U2D
VDD
VDD
VBAT
VDDA
VSS (ExPAD)
C21
100n
C22
100n
C205
10u
C14
100n
Connectors: Breakout and SWD
SWD Connector
I2C, UART and GPIO Breakout Connector
J2A
J2B
FTSH-105-01-L-DV-K-TR
FTSH-105-01-L-DV-K-TR
NRST
P1
62200821121
Title
Size
A4
Number
Revision
Date: 8/12/2026
Sheet of
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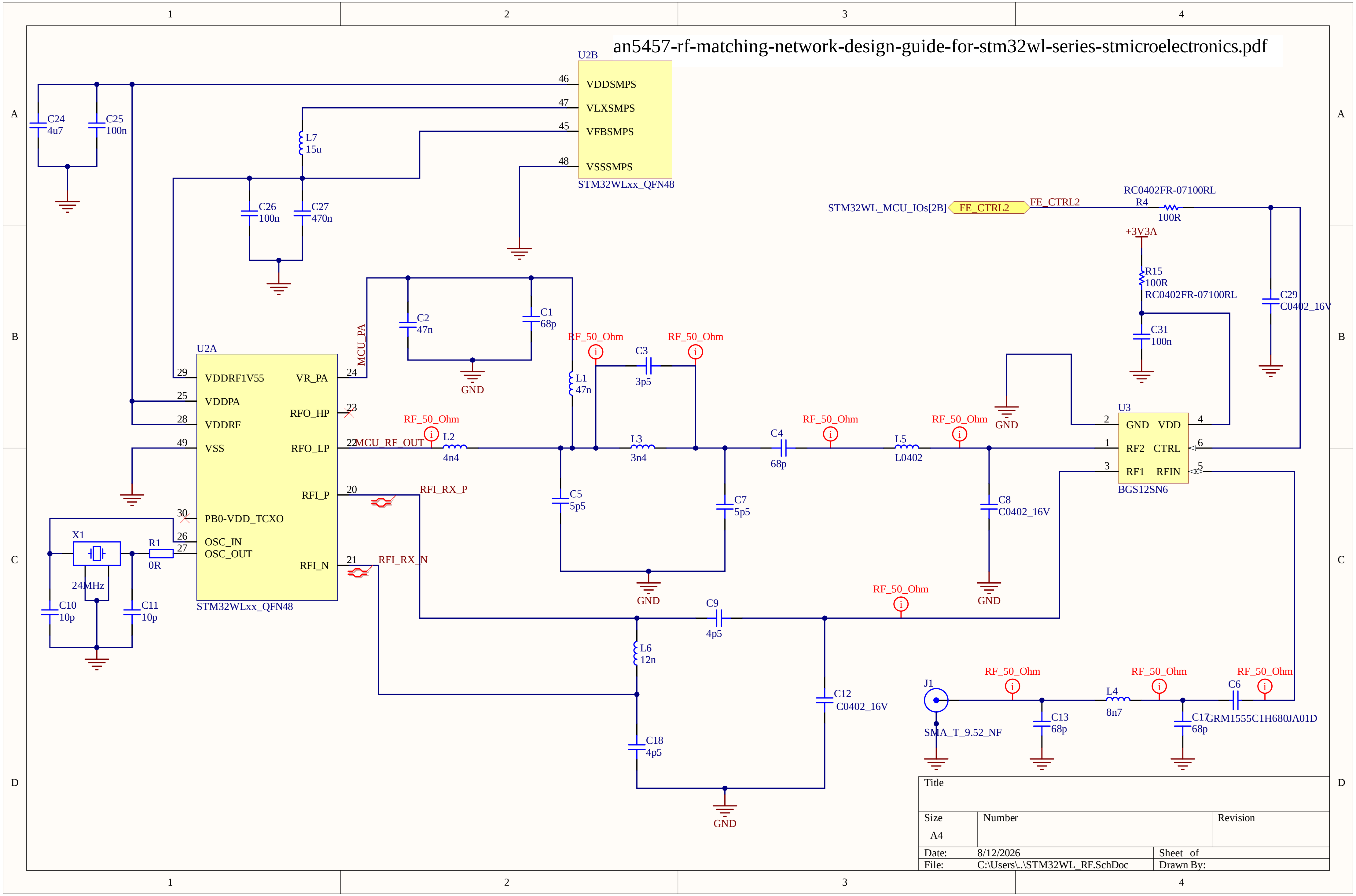
Drawn By:


an5457-rf-matching-network-design-guide-for-stm32wl-series-stmicroelectronics.pdf
U2B
VDDSMPS
VLXSMPS
VFBSMPS
VSSSMPS
STM32WLxx_QFN48
C24 4u7
C25 100n
L7 15u
C26 100n
C27 470n
U2A
VDDRF1V55
VDDPA
VDDRF
VSS
PB0-VDD_TCXO
OSC_IN
OSC_OUT
VR_PA
RFO_HP
RFO_LP
RFI_P
RFI_N
STM32WLxx_QFN48
MCU_PA
MCU_RF_OUT
RFI_RX_P
RFI_RX_N
X1 24MHz
R1 0R
C10 10p
C11 10p
C2 47n
C1 68p
GND
RF_50_Ohm
L1 47n
L2 4n4
C3 3p5
L3 3n4
C5 5p5
C7 5p5
C4 68p
L5 L0402
C8 C0402_16V
L6 12n
C9 4p5
C12 C0402_16V
C18 4p5
STM32WL_MCU_IOs[2B]
FE_CTRL2
RC0402FR-07100RL
R4 100R
+3V3A
R15 100R
RC0402FR-07100RL
C31 100n
C29 C0402_16V
U3
GND VDD
RF2 CTRL
RF1 RFIN
BGS12SN6
J1
SMA_T_9.52_NF
C13 68p
L4 8n7
C17 68p
C6 GRM1555C1H680JA01D
Title
Size A4
Number
Revision
Date: 8/12/2026
File: C:\Users\..\STM32WL_RF.SchDoc
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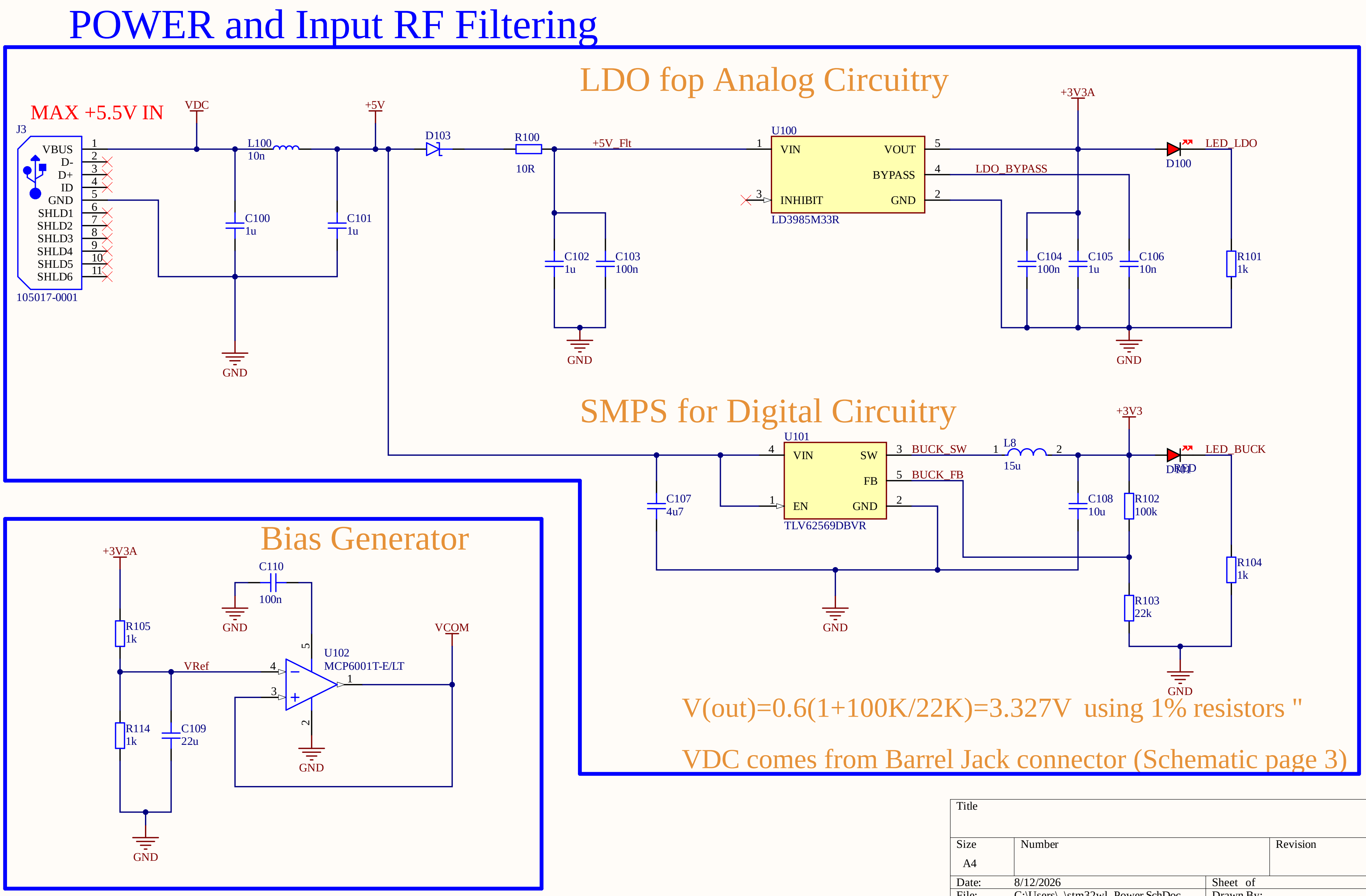
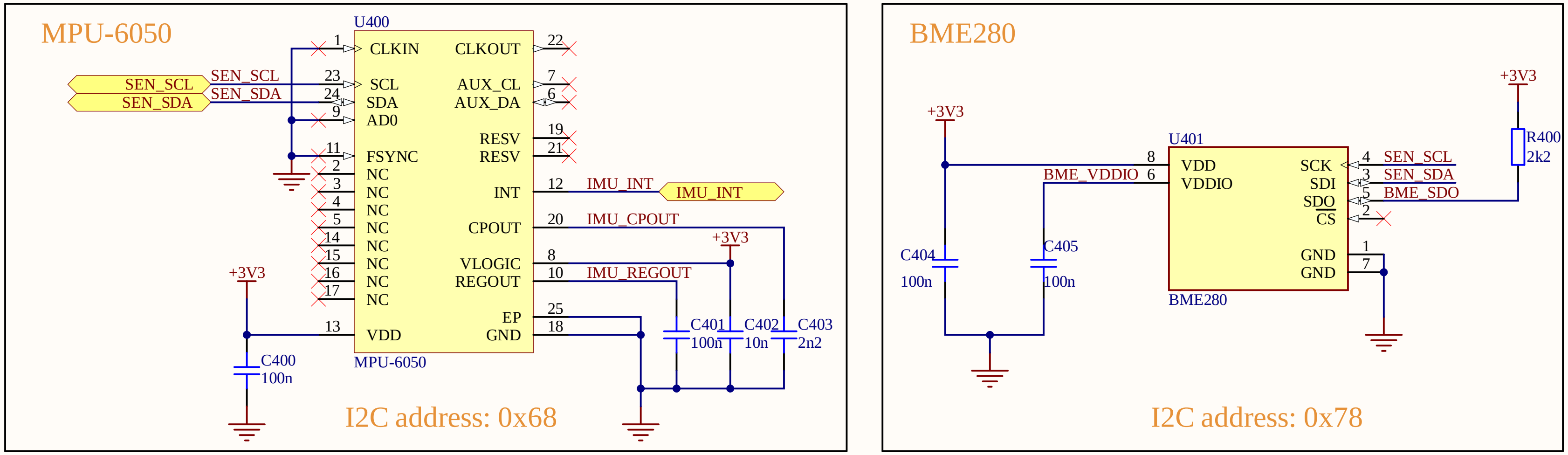


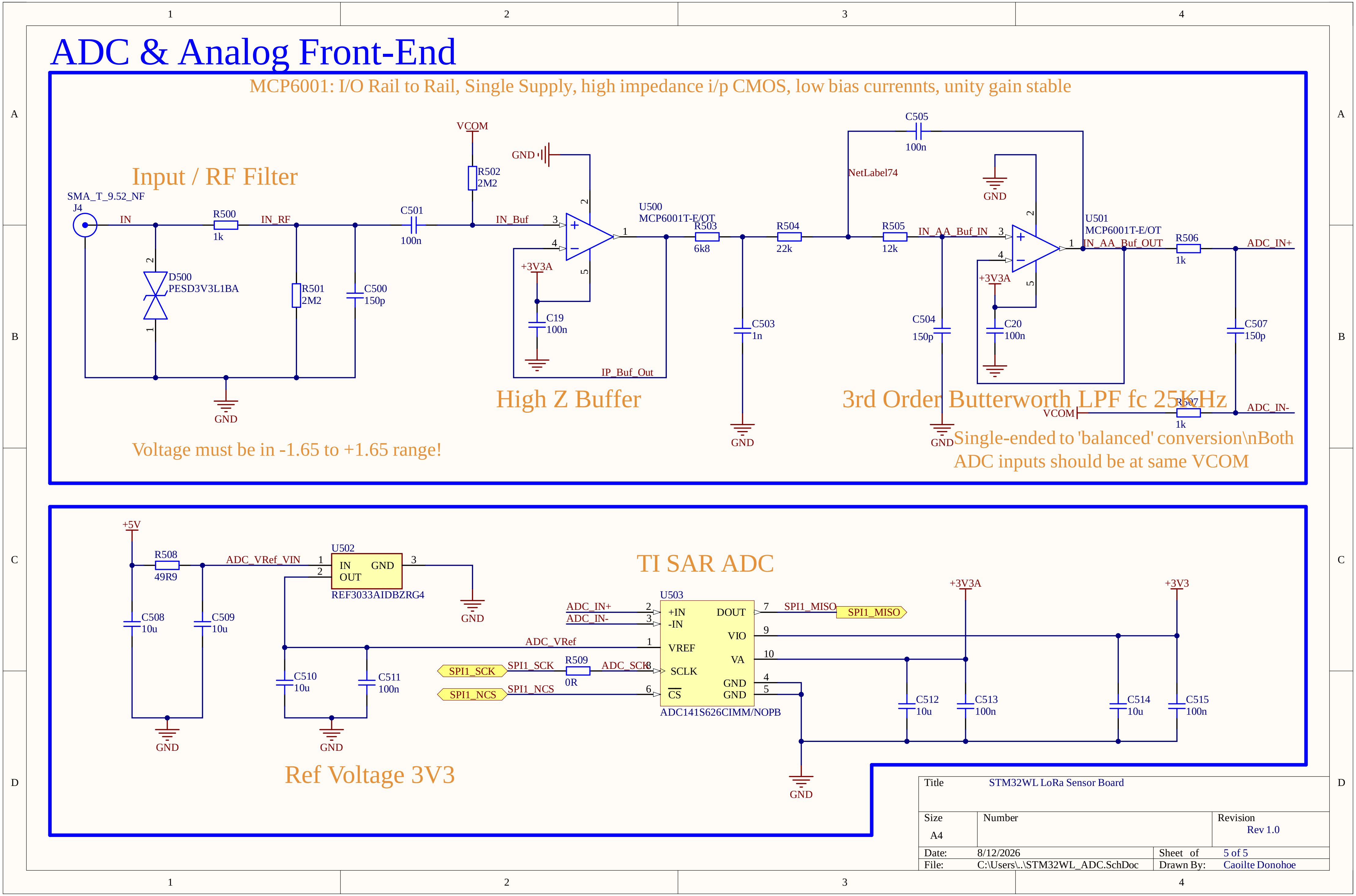
Title		
Size: A4	Number	Revision
Date: 8/12/2026		Sheet of
File: C:\Users\..\stm32wl_Power.SchDoc		Drawn By:

PCB Mounted Sensors

Inertial Measurement Unit (MPU-6050) and BME280 (Temp/Humidity Pressure)



Title	STM32WL LoRa Sensor Board		
Size: A4	Number		Revision: Rev 1.0
Date:	8/12/2026	Sheet of	5 of 5
File:	C:\Users\..\STM32WL_Sensors.SchDoc	Drawn By:	Caoilte Donohoe


ADC & Analog Front-End
MCP6001: I/O Rail to Rail, Single Supply, high impedance i/p CMOS, low bias currennts, unity gain stable
Input / RF Filter
SMA_T_9.52_NF
J4
IN
R500
1k
IN_RF
D500
PESD3V3L1BA
R501
2M2
C500
150p
C501
100n
VCOM
R502
2M2
IN_Buf
GND
U500
MCP6001T-E/OT
+3V3A
C19
100n
IP_Buf_Out
High Z Buffer
R503
6k8
R504
22k
C503
1n
C505
100n
NetLabel74
R505
12k
IN_AA_Buf_IN
C504
150p
U501
MCP6001T-E/OT
IN_AA_Buf_OUT
C20
100n
R506
1k
ADC_IN+
C507
150p
R507
1k
ADC_IN-
3rd Order Butterworth LPF fc 25KHz
Voltage must be in -1.65 to +1.65 range!
Single-ended to 'balanced' conversion\nBoth ADC inputs should be at same VCOM
+5V
R508
49R9
ADC_VRef_VIN
U502
IN GND
OUT
REF3033AIDBZRG4
C508
10u
C509
10u
C510
10u
C511
100n
ADC_VRef
Ref Voltage 3V3
TI SAR ADC
U503
+IN
-IN
VREF
SCLK
CS
DOUT
VIO
VA
GND
GND
ADC141S626CIMM/NOPB
SPI1_MISO
SPI1_SCK
SPI1_NCS
R509
0R
ADC_SCK
+3V3A
+3V3
C512
10u
C513
100n
C514
10u
C515
100n
Title STM32WL LoRa Sensor Board
Size A4
Number
Revision Rev 1.0
Date: 8/12/2026
File: C:\Users\..\STM32WL_ADC.SchDoc
Sheet of 5 of 5
Drawn By: Caoilte Donohoe